

# PIYUSH LOKHANDE

LeetCode | Codeforces | CodeChef |  GitHub |  LinkedIn |  Email |  +91-9623990067

## EDUCATION

National Institute of Technology Raipur

*B.Tech in Computer Science and Engineering — CPI :8.34*

Aug 2023 – Present

*Raipur, Chhattisgarh*

Ad. S. G. Rathod Vidyalaya, Lakh (Khind), Darwha

*Class 12th HSC — 83.17%*

Aug 2020 – June 2022

*Darwha, Maharashtra*

## TECHNICAL SKILLS

- **Languages:** C++, Go, JavaScript, Python
- **Web & Backend:** Go (Golang), Node.js, Express.js, REST APIs, React
- **Databases & Tools:** PostgreSQL, MongoDB, Redis, Docker, Linux, Git

## COURSEWORK

- Data Structures & Algorithms
- Object-Oriented Programming (OOP)
- Web Development
- Computer Networks

## PROJECTS

Distributed Job Queue System 

2026

- Designed and implemented a production-oriented distributed job queue for asynchronous task processing using **Go** and **Redis**.
- Built concurrent worker pools with goroutines and channels, supporting **graceful shutdown** via context and OS signals.
- Implemented reliability mechanisms including retry with exponential backoff and **backpressure-based** load protection.
- Added observability through **structured logging** and a /metrics endpoint; containerized using **Docker** and Docker Compose.
- Tech Stack: **Go**, **Redis**, **Docker**, net/http, Goroutines, Channels

Gatekeeper-Go (High-Performance API Gateway) 

2026

- Built a low-latency proxy engine in **Go** using **httputil.ReverseProxy** for automated path-rewriting and routing.
- Created a high-throughput load balancer using hardware-level **sync/atomic** math to avoid concurrent race conditions.
- Developed a distributed rate limiter via a Token Bucket **Redis Lua script** to eliminate read-write race conditions at the edge.
- Secured endpoints via **RS256 JWT validation**, offloading signature decryption and propagating context headers downstream.
- Injected an **X-Correlation-ID** middleware pipeline to enforce structured JSON distributed tracing across services.
- Tech Stack: **Go**, **Redis**, **Docker**, **JWT (RS256)**, **Concurrency (sync/atomic)**, **Lua**

BP Guardian  

2025

- Built a secure healthcare platform for patient–doctor interaction with blood pressure monitoring and protected data workflows.
- Implemented **OTP-based email verification** and role-based access control to safeguard sensitive medical records.
- Designed and trained a **Random Forest** ML model to suggest medicines from BP risk indicators, with Gemini AI providing general health advice and contextual guidance.
- Tech Stack: **Node.js**, **Express.js**, **MongoDB**, **React.js**, **Random Forest**, Gemini AI

## ACHIEVEMENTS

- **LeetCode** – Max Rating **1870**
- **CodeChef** – Max Rating **1778**
- Solved **1000+** algorithmic and data structures problems across competitive programming platforms
- Achieved **Global Rank 48** in **CodeChef Starters 180** 

## POSITION OF RESPONSIBILITY

Core Coordinator - Web Domain

*ACE (Association of Computer Engineers)*

2025–Present

*NIT Raipur*

- Contributed to the development of the official ACE website, supporting event coordination and resource management for 1,000+ engineering students.